Data Science and AI for Finance
DA6701

Session 2 – Basic Financial Concepts

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Basic financial concepts

- Basics of compounding, Log returns, stock price distribution.
- Efficient Market Hypothesis, No Arbitrage Principle, and one-step binomial tree
- Risk and expected return (risk free + risky asset; two risky assets)

Time value of money and continuous compounding

- The compound interest formula is $\mathrm{V(t)} = \mathrm{P}(1+\frac{\mathrm{r}}{\mathrm{m}})^{\mathrm{tm}}$
- It is evident that if we increase r, t or P that V(t) increases.
- What happens if we increase m.
- The P term is the same, t is the same so the question is for m < k is $(1+\frac{\mathrm{r}}{\mathrm{m}})^{\mathrm{m}} < (1+\frac{\mathrm{r}}{\mathrm{k}})^{\mathrm{k}}$?
- Formula for Binomial expansion is $(a+b)^n = \sum_{j=0}^{n} \binom{n}{j} a^{n-j} b^j$
- In our case this becomes $\left(1+\frac{r}{m}\right)^m = 1 + r + \frac{1}{2!} r^2 \left(1-\frac{1}{m}\right)$
- $+\frac{1}{3!} r^3 \left(1-\frac{1}{m}\right)\left(1-\frac{2}{m}\right) + \cdots + \frac{1}{m!} r^m \prod_{i=1}^{m-1} \left(1-\frac{i}{m}\right)$.
- So what happens if k>m. Each term after the second term and the total number of terms?

Continuous compounding and defining returns

- WKT $\lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n = e^x$
- Therefore $\mathrm{V(t)} = \mathrm{P.e^{rt}}$
- This has implications on how we look at returns. Tradition growth per unit time step is $\mathrm{p_2 = p_1(1+r_1)}$, Therefore $\mathrm{r_1}$ was $\frac{\mathrm{p_2}}{\mathrm{p_1}} - 1$. Whereas when $\mathrm{p_2 = p_1\ e^{r_1}}$, then $\mathrm{r_1} = \ln(\frac{\mathrm{p_2}}{\mathrm{p_1}})$
- Continuous compounding not only reflects growth in a constantly evolving real-world but is also convenient. If we had another period $\mathrm{p_3 = p_2(1+r_2)}$. Then returns over two time periods defined by $\frac{\mathrm{p_3}}{\mathrm{p_1}} - 1 = (1+\mathrm{r_1})(1+\mathrm{r_2}) - 1$. Which is multiplicative.

Continuous compounding and defining returns

- Log returns are entirely additive. $\ln(\frac{p_2}{p_0}) = \ln(\frac{p_2}{p_1} x \frac{p_1}{p_0}) = \ln(\frac{p_2}{p_1}) + \ln(\frac{p_1}{p_0})$. This is not only convenient but more intuitive.
- **Scenario:**

Stock starts at $100.

Day 1: Goes up to $150.

Day 2: Goes back down to $100.

- **The Simple Return View (Deceptive):**

Day 1: (150−100)/100=+50%

Day 2: (100−150)/150=−33.3%

Average Return: (50−33.3)/2=+8.35%

- *Wait... the stock is back to $100. Why is the average positive? This implies you made money. You didn't.*
- **The Log Return View (Honest):**

Day 1: ln(150/100)=+0.405

Day 2: ln(100/150)=−0.405

- **Total Return:** 0.405+(−0.405)=0

Log returns to stock prices

- When we apply this principle to stock prices, over some reasonable period of time then we can express its price change as
- $\ln\left(\frac{P_T}{P_0}\right) = \underbrace{\ln\left(\frac{P_1}{P_0}\right)}_{r_1} + \underbrace{\ln\left(\frac{P_2}{P_1}\right)}_{r_2} + \cdots + \underbrace{\ln\left(\frac{P_T}{P_{T-1}}\right)}_{r_n}$
- Then as long as each return in approximately Normal or 'n' is large enough for CLT to take effect we can reason that $\ln\left(\frac{P_T}{P_0}\right) \sim Normal$
- This can be expressed as $\ln(P_T) - \ln(P_0) \sim Normal$, which implies that $\ln(P_T) \sim Normal$.
- A log-normal distribution is defined as $X = e^{\mu+\sigma Z}$, where Z is a standard Normal.
- By definition therefore $P_T \sim LogNormal$

Efficient market hypothesis and no-arbitrage principle.

- EMH states that transparent markets (asset prices) reflect all available information.
- Implication: If a "Money Machine" (arbitrage strategies or positive theoretical alpha) appeared, market participants would exploit it with infinite capital until prices moved and the opportunity vanished.
- Side note: What is arbitrage? It is a riskless profit with zero net investment.
- Side note: What is alpha? It defines how much money you made **beyond** what you *should* have made given the risk you took (will revisit in CAPM). Alpha is not just beating the market. It is $\mathrm{E}(\mathrm{R}_{\mathrm{actual}})$-$\mathrm{E}(\mathrm{R}_{\mathrm{i}})$where $\mathrm{E}(\mathrm{R}_{\mathrm{i}}) = \mathrm{R}_{\mathrm{f}} + \beta_i(\mathrm{E}(\mathrm{R}_{\mathrm{market}}) - \mathrm{R}_{\mathrm{f}})$; where

$$\beta_i = \rho_{i,\mathrm{market}} \cdot \frac{\sigma_i}{\sigma_{\mathrm{market}}}$$

- What do you get for value $\beta_i = 0,1,2$ (what about negative values)

Efficient market hypothesis and no-arbitrage principle.

- Forms of EMH:
  - Weak form: Prices reflect past information (technical analysis ineffective). (next 2 slides)
  - Semi-strong form: Prices reflect all public information (fundamental analysis ineffective).(claims alpha should not be possible, but debatable).
  - Strong form: Prices reflect all information, public and private (even insider info ineffective).
- Does EMH hold? Circular reasoning: if EMH were 100% true (all the time) there should be no arbitrageurs, traders and buy-side prop desks. But it is because these stakeholders exist, EMH must be true.
- Bottom line: EMH motivates the idea that arbitrage opportunities (riskless profit) should not **persist** — leading us to the **no-arbitrage principle**.

Weak EMH and Technical analysis

- As per even Weak EMH Technical Analysis – specifically, retail technical analysis is mathematically impossible.
- Things like 50 day MA crossing 200 day MA, support lines, Bollinger bands, Elliot waves, lack even a shred of statistical scrutiny.
- It supports Pareidolia, serves as a self fulfilling prophecy, might support mob effects, provide commission (and product sale for sell side).
- However, when a desk at Goldman says they are trading technicals they aren’t looking at funny shapes on graphs but microstructure and flow (like orderbook imbalance, mean reversion, gamma exposure)

Weak EMH and Technical Analysis

- **Burton Malkiel**, Princeton economist and author of *A Random Walk Down Wall Street*, is the most aggressive academic critic of technicals.
  - "Financial forecasting appears to be a science that makes **astrology** look respectable.“;.
- **Warren Buffett**
  - "I realized that technical analysis didn't work when I **turned the chart upside down and didn't get a different answer**."
- **Peter Lynch**, who managed the Magellan Fund to 29% average annual returns (beating the market for 13 years straight)
  - "Charts are great for **predicting the past**."

No Arbitrage Principle

- **The Setup:** Consider two assets in our economy:
- **Risk-Free Asset (B):** You deposit $1 into a bank. At time T, it grows to (1+r). This return is guaranteed.
- **Risky Asset (S):** You buy a Stock for S_0. At time T, its value S_T is uncertain.
- **The Definition of Arbitrage:** An arbitrage opportunity is a strategy where:
- You invest **$0** today (Net value of portfolio at t=0 is zero).
- You have **zero probability** of losing money at t=T.
- You have a **positive probability** of making money at t=T.
- **The Equation (Dominance):** If we have two assets that are guaranteed to produce the exact same return, they must have the same price. If Risk-Free Asset A pays $110, and Risk-Free Asset B pays $110, then:
- Price(A)=Price(B)

No Arbitrage Principle

- What does violation of Arbitrage look like?
  - **Bank Rate:** 5% (Guaranteed).
  - **Stock:** Currently trading at $100. A "Guru" guarantees the stock will be worth exactly $110 in one year (10% return) with zero uncertainty.
  - **The Arbitrage:**
    1. Borrow $100 from the bank at 5%. (Cost in 1 year = $105).
    2. Buy the Stock at $100.
    3. Wait 1 year. Sell Stock for $110. Repay Bank $105.
    4. **Profit:** $5.00 with $0 investment and zero risk.
  - **Market Correction:** Everyone buys the stock. S_0 rises to $104.76 (where $104.76 x 1.05 = 110$). The free profit vanishes.
  - The role of Derivatives for more realistic examples of Arbitrage.

One step Binomial tree

- **The "No-Arbitrage Condition" Theorem:** For this market to be valid (no free money), the Risk-Free return (R = 1+r) must lie strictly *between* the Down return (d) and the Up return (u) : ***d<(1+r)<u***

- Proof by Contradiction (The Arbitrage Logic):

Case 1: The Stock is "Too Good" ($(1+r) < d < u$)

•*Scenario:* Bank pays 2% (R=1.02). Stock Worst Case is +5% (d=1.05).

•*Strategy:* Borrow from Bank. Buy Stock.

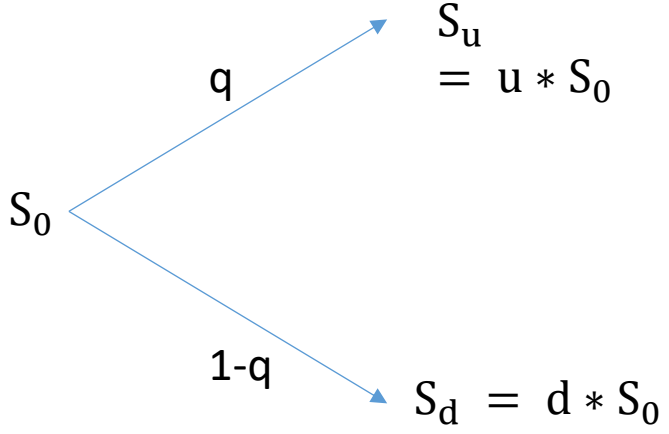
•*Outcome:* Even in the worst case, the Stock pays 5%, which covers the 2% loan. You keep the difference risk-free. **Arbitrage.**

Case 2: The Stock is "Too Bad" ($d < u < (1+r)$)

•*Scenario:* Bank pays 10% (R=1.10). Stock Best Case is +5% ($u=1.05$).

•*Strategy:* Short Sell the Stock. Invest proceeds in Bank.

•*Outcome:* The Bank interest (10%) will always be enough to buy back the stock (max 5% growth) and leave profit. **Arbitrage.**



One step Binomial tree

- **The Risk-Neutral Probability (q)** Since d<1+r<u, we can rearrange the terms to find a specific "weight" or probability (q) that equates the Stock's expected return to the Risk-Free rate.

$$S_0 = \frac{1}{1+r}[q.S_u + (1-q)S_d]$$

$$q = \frac{(1+r)-d}{u-d}$$

- **Numerical Example:**

S_0 =100

u=1.2 (S_u =120)

d=0.85 (S_d =85)

r=0.05 (R=1.05)

- **Check Condition:** 0.9<1.05<1.2. (Valid Market).
- **Calculate q:**

$$q = \frac{1.05-0.85}{1.2-0.85} \cong 0.5714$$

- Do we have an arbitrage opportunity if the true q was actually 0.8 and we knew that? No.

Risk and expected Return

- Two assets: Risky and Risk Free assets:
    1. Risk-Free Asset (F): Guaranteed return, zero variance.
    2. Risky Portfolio (P): A basket of stocks/bonds with uncertain return.
- The Expected return of these assets
    1. r_f is not a random variable. Therefore $r_f = E(r_f)$
    2. r_p is a random variable with expected return = $E(r_p)$The Expected return of these assets
- Uncertainty in the return of the assets
    1. r_f is not a random variable. Therefore $\sigma(r_f) = 0$
    2. r_p is a random variable. We express $\sigma_p = \sigma(r_p)$
    3. COV $(r_f, r_p) = 0$
- 'C' is a portfolio where allocate y (a weightage between 0 to 1) in the risky asset and (1-y) in the risk free asset.

Risk and expected return

- r_c is the return on the portfolio. We can see that
  - $E(r_c) = y \cdot E(r_p) + (1 - y) \cdot r_f$
  - $\sigma_c = \sqrt{y^2 \sigma_p^2} = y \cdot \sigma_p$

•We can combine the Return and Risk equations to eliminate y. This gives us the equation of a straight line:

•$E(r_c) = r_f + [\frac{E(r_p) - r_f}{\sigma_p}].\sigma_c$

•The Intercept: rf (The reward for waiting).

•The Slope (S): The Sharpe Ratio.

•$s = [\frac{E(r_p) - r_f}{\sigma_p}]$

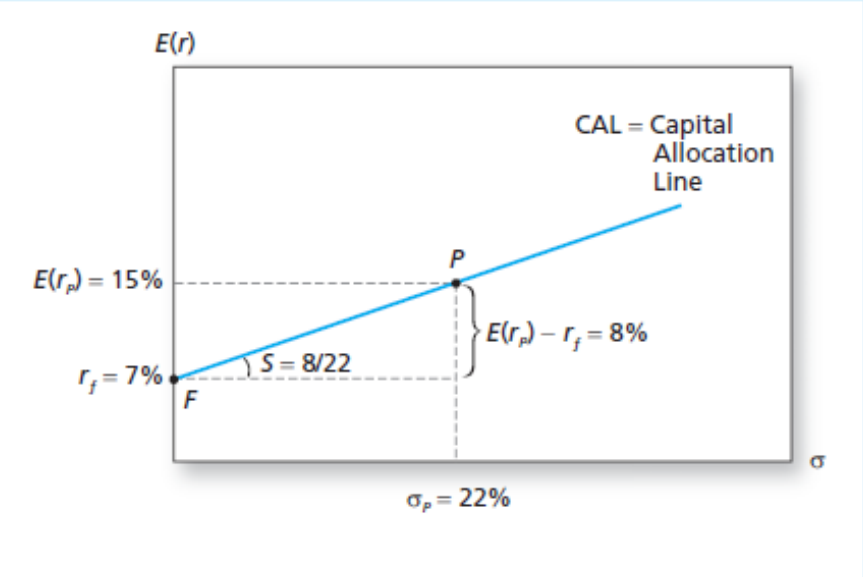


Figure 6.4 The investment opportunity set with a risky asset and a risk-free asset in the expected return–standard deviation plane

Bodie, Zvi, Alex Kane, and Alan Marcus. *Ebook: Essentials of investments: Global edition*. McGraw Hill, 2013.

Interpretation: The "Reward-to-Volatility" ratio. How much extra return do you get for every unit of risk?

Risk and expected return

- How do you combine risk and return. How do you capture risk aversion
  - Utility function: $U = E(\mathrm{r_c}) - \frac{1}{2} A \cdot {\sigma_\mathrm{c}}^2$
  - A = risk aversion coefficient
  - Higher A → steeper indifference curves → less risky allocation.
- Maximize Utility
  - $\frac{dU}{dy} = \left(E\left(r_p\right) - r_f\right) - A \cdot y \cdot \sigma_p^2 = 0$
- Solution
  - $y^* = \frac{E\left(r_p\right) - r_f}{A \cdot \sigma_p^2}$

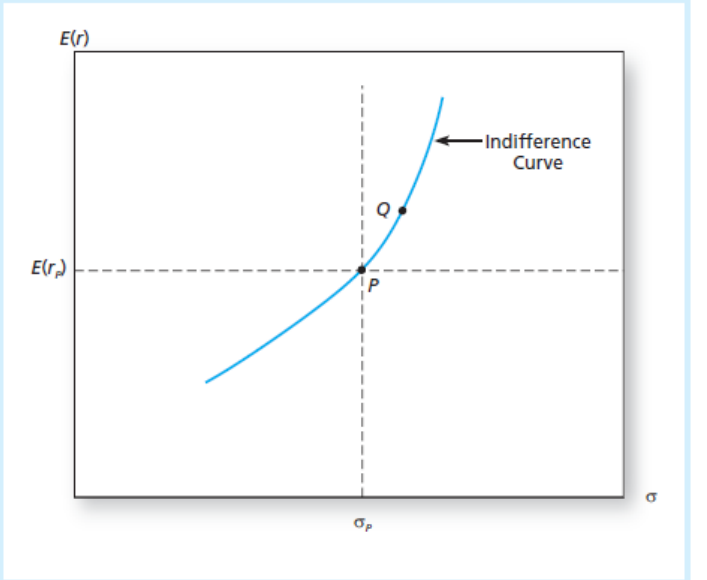


Figure 6.2 The indifference curve

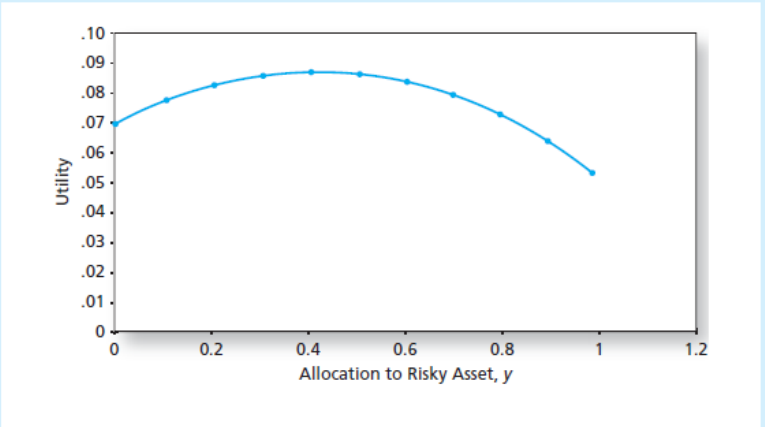


Figure 6.6 Utility as a function of allocation to the risky asset, *y*

Bodie, Zvi, Alex Kane, and Alan Marcus. *Ebook: Essentials of investments: Global edition*. McGraw Hill, 2013.

Risk and Expected Return

- Two risky assets:
  - We could find the minimum variance portfolio
  - We could find the portfolio that maximizes the Sharpe ratio.
- Minimum Variance Portfolio:

$$\sigma_p^2 = w^2\sigma_1^2 + (1-w)^2\sigma_2^2 + 2w(1-w)\rho_{12}\sigma_1\sigma_2$$

$$\frac{d\sigma_p^2}{dw} = 2w\sigma_1^2 - 2(1-w)\sigma_2^2 + 2(1-2w)\sigma_{12}$$

$$0 = 2w(\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}) + (2\sigma_{12} - 2\sigma_2^2)$$

$$w^* = \frac{\sigma_2^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}$$

Risk and expected return

- Implications on diversification. Imagine if $\sigma_2 < \sigma_1$, then we can easily find scenarios where w should be 0 or negative (short sell security 1 and go leveraged on security 2). But is there a scenario where we want a w>0?
- If we take the denominator portion alone we can rewrite it as:

$$D = (\sigma_1 - \sigma_2)^2 + 2\sigma_1\sigma_2(1 - \rho).$$

D has to be positive

Therefore if $\sigma_2^2 - \sigma_{12} > 0$, then w>0

Which means $\sigma_2^2 > \rho\sigma_1\sigma_2$ or in other words $\rho < \frac{\sigma_1}{\sigma_2}$

Risk and expected return

- Portfolio that maximizes Sharpe ratio
- Formulation:

$$S(w) = \frac{E(R_p) - R_f}{\sigma_p} = \frac{w\big(E(R_1) - R_f\big) + (1-w)\big(E(R_2) - R_f\big)}{\sqrt{w^2\sigma_1^2 + (1-w)^2\sigma_2^2 + 2w(1-w)\sigma_{12}}}$$

- Won't do the steps, but direction:
  - Differentiate this w.r.t w
  - Use the quotient rule to realize that maximum occurs when u'/u = v'/v
  - Chain rule + Lots of algebra

$$w_1^* = \frac{\big(E(R_1) - R_f\big)(\sigma_2^2 - \rho\sigma_1\sigma_2); -; \big(E(R_2) - R_f\big)(\sigma_{12} - \sigma_1^2)}{\big(E(R_1) - E(R_2)\big)(\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2)}$$